

Mathematics 2025/26, Homework 3
bonus: 2 points, deadline: April 13, 2026

1. Solve the equations

(a) $\ln x = 8$

(b) $e^x = 5$

2. Write the definition of the derivative of a function f at the point x_0 .

3. Find derivatives of the following functions

(a) $y = x^7 - 2x^5 + 5x^4 + x^3 - 7x + 1$

(b) $y = \frac{1}{x^2} - \frac{1}{\sqrt{x}}$

(c) $y = (x^3 + 1) \ln x$

(d) $y = \frac{2x}{x^3 + 1}$

(e) $y = \sin(2x + 5)$

(f) $y = (x^2 - x + 1)^7$

Instructions for writing homework:

- Write your homework with solution (not only the results).
- Take a photo of the homework and convert the picture to PDF (use <https://tools.pdf24.org/en/jpg-to-pdf>). Alternatively, use a different way to make a PDF file with your homework.
- Compress the file if it is large (use <https://tools.pdf24.org/en/compress-pdf>).
- Upload the final PDF file to “cousework submissions” in UIS.